

Sign and Like-Term Errors in Polynomial Operations

The three errors this sheet hunts for:

- **The stranded minus.** A subtraction sign in front of a parenthesis multiplies *every* term inside it, not only the first one.
- **The false like term.** Terms combine only when their variable parts are identical: x^2 and x never merge, however tempting the coefficients look.
- **The missing middle term.** Squaring a binomial is a product, so the two cross terms survive; $(a - b)^2$ is not $a^2 + b^2$.

On every find-and-fix item, quote the exact step that goes wrong, write the correct simplification, and then confirm it by substituting the value the problem names. A substitution check is the fastest way to prove a sign is wrong without re-reading the whole line.

1. Combine like terms: $7x^2 + 3x - 5x^2 + 8x - 2$.

Answer: _____

2. Subtract, showing the line in which the subtraction sign has been distributed: $(5x^2 - 3x + 4) - (2x^2 + 7x - 6)$.

Answer: _____

3. Multiply and simplify: $-3x(2x^2 - 4x + 5)$.

Answer: _____

4. Find and fix the mistake. A student writes

$$(4x^2 - 6x + 1) - (x^2 - 2x - 5) = 3x^2 - 8x - 4$$

Name the step that goes wrong, write the correct simplification, then evaluate your corrected answer at $x = 2$ to confirm it.

Answer: _____

5. Find and fix the mistake. A student writes

$$-2x(3x - 7) + 5(x - 4) = -6x^2 - 9x - 20$$

Exactly one term is mis-signed. Name it, expand the left side correctly, then evaluate your corrected answer at $x = 1$ to confirm it.

Answer: _____

6. Factor $x^2 - 5x - 24$. Then check the factorisation by evaluating your factored form at $x = 1$ and comparing it with the value of $x^2 - 5x - 24$ at $x = 1$.

Answer: _____

7. Find and fix the mistake. A student claims that $(2x - 3)^2$ and $4x^2 + 9$ are the same expression. Decide whether they are, expand $(2x - 3)^2$ correctly, and support your decision by evaluating your expansion at $x = 1$.

Answer: _____

8. Explain. A classmate says: “When you subtract polynomials you only have to flip the sign of the first term inside the parentheses, and anyway $3x^2 + 5x$ is just $8x^2$.” Write a short paragraph correcting both claims. State the subtraction rule properly, give one substitution that proves $3x^2 + 5x$ and $8x^2$ are different, and say what has to be true of two terms before they may be combined.